Shashi Bhushan Vijay

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SUMMARY

M.Tech Computer Science Engineering student and patent holder specializing in ML pipelines, distributed systems, and backend engineering. Proven track record of shipping production-grade systems - including a patented indoor positioning solution with $R^2 = 0.978$ accuracy. Experienced across the full stack, from edge hardware integration to LLM-powered SaaS platforms.

EXPERIENCE & PROJECTS

AutoML Assistant *Jan 2026*
Project *Repo | Live Demo*

- Architected an end-to-end AutoML platform using Streamlit + Python to automate data upload, preprocessing, analysis, model training, and optimization workflows.
- Designed a modular ML backend supporting **regression, classification, and time-series forecasting** with algorithms like XGBoost, Random Forest, ARIMA/SARIMAX, and optional LSTM.
- Integrated LLM-powered model recommendation (RAG) using LangChain, Groq (Llama 3.1), and FAISS to suggest suitable models based on dataset statistics and ML best practices.
- Implemented **SHAP-based explainability** (feature importance, beeswarm, waterfall, dependence plots) to make model predictions interpretable for end users.
- Exposed trained models through a **FastAPI** inference API (`/predict`, `/predict/csv`) with **persistent workspace/model management** for reusable deployment-ready ML pipelines.

Patent: Coordinated Indoor Position Determination via Multi-Stage Signal Processing *Dec 2024–Dec 2025*
VIT, Vellore, IN *Application No. 202541115892*

- Delivered an indoor localization system using Wi-Fi RSSI fingerprinting and Bluetooth integration, achieving approx. **1.6 m positioning accuracy (MAE: 1.468 m, R^2: 0.978)** in complex environments.
- Deployed an ESP32-based autonomous Wi-Fi sniffer on a robotic car, automating radio-map generation and capturing **4,277 RSSI records across 209 access points** - eliminating manual site surveys.
- Designed and optimized a Dynamic Stacking ensemble (KNN, Random Forest, XGBoost + Gradient Boosting meta-learner), cutting localization error by **approx. 43% (RMSE: 3.124 to 1.787)**.
- Engineered a real-time data pipeline using MQTT, Raspberry Pi, and Python, transforming raw Wi-Fi scans into machine-learning-ready datasets for live location predictions.
- Implemented an **A*-based** indoor navigation module using NetworkX for optimal route generation and dynamic path recalculation from predicted user locations to destinations.

RiskLens — Credit & Fraud Risk Analytics Platform
Python · SQL · PostgreSQL · XGBoost · SHAP · Streamlit · Plotly

- Architected RiskLens, an end-to-end credit and fraud risk analytics platform in Python and SQL (PostgreSQL), combining an 8-rule deterministic domain engine (`VELOCITY_SURGE`, `UNUSUAL_LOCATION`) with an XGBoost ML pipeline to intercept and score **590,540 real-world transactions** across the customer lifecycle.
- Engineered robust ETL data stewardship and validation pipelines (`DataValidator`), enforcing schema integrity, handling missing/duplicate attributes, and automating **data quality scoring (100/100)** prior to database ingestion and model training.
- Optimized ML performance and resource efficiency via stateful chronological feature engineering (**29 leakage-safe behavioral features**) and categorical cardinality capping, reducing dense feature matrix RAM by **99.5% (43.5 GiB to 218 MB)** without sacrificing accuracy.
- Formulated a custom business utility optimization strategy (`ThresholdOptimizer`) selecting an optimal **0.3500 decision cutoff** to achieve **66.98% fraud recall (0.7607 ROC-AUC)**, with real-time SHAP feature attribution (`shap.TreeExplainer`) for regulatory explainability.
- Developed an interactive full-stack Streamlit & Plotly dashboard for live fraud alert investigation and automated model performance monitoring via Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) statistical **data drift detection**.

FLEX-DCA AI Platform - AI-Powered Debt Collection Allocation & Compliance System *Jan–Feb 2026*
Project *Repo*

- Engineered a full-stack debt recovery platform using FastAPI, React, and SQLite to ingest cases, run predictions, auto-allocate DCAs, and track operational KPIs via REST APIs.
- Trained and integrated **3 CatBoost models** (recovery probability, aging risk, closure speed) with financial feature engineering for real-time decision support.
- Implemented a **RAG-based** legal copilot using **FAISS** + **SentenceTransformers** + **Groq LLaMA 3.1**, enabling policy-grounded answers for FDCPA, Regulation F, and RBI compliance queries.
- Created an operations dashboard with KPI monitoring and case workflows, improving visibility into recovered/open cases and allocation outcomes.
- Structured backend into modular services (ML, allocation, RAG, dashboard, ingestion) to support scalable, production-style architecture for enterprise debt operations.

EDUCATION

Vellore Institute of Technology, Vellore *Expected April 2027*
Integrated M.Tech in Computer Science Engineering GPA: 7.0/10
Relevant Coursework: Machine Learning, Distributed Systems, Data Structures & Algorithms, Database Management, Computer Networks, Operating Systems, Cloud Computing

SM Public School, Haridwar *2020–21*
Junior College (Class XII) 91.2%

SM Public School, Haridwar *2018–19*
High School (Class X) 87%

TECHNICAL SKILLS

Languages & Core: Python, C++, Java, JavaScript, SQL
Backend & Systems: FastAPI, REST APIs, Microservices, MQTT, Docker, CI/CD
Databases: PostgreSQL, SQLite, Redis | **Tools & Cloud:** Git, AWS, Streamlit, Power BI
ML & AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, LSTM, SHAP, RAG, FAISS, LangChain, Groq

LEADERSHIP & EXTRACURRICULARS

Sponsorship Coordinator, ISTE Chapter – VIT Vellore

- Secured **Rs. 1,00,000+ in corporate sponsorships** for flagship hackathon (Horizon) by pitching to **100+ founders and startups**; orchestrated end-to-end event logistics and developed outreach frameworks that established strategic industry connections for future initiatives.

ACHIEVEMENTS

1st Place, ARC Hackathon (ARC Lab, VIT) – Created a GUI-based AutoML tool that automatically benchmarks and recommends the best ML model for any given dataset, winning against competing teams.

CERTIFICATIONS

Stock Market Basics & Technical Analysis – View Certificate
Designing Seamless Notification Service – View Certificate